Based on the workflow and technical components outlined in the PDFs, here is a recommended 8-week timeline to get you from the planning stage to a functional Minimum Viable Product (MVP). This timeline assumes a dedicated, part-time effort from both of you.

Phase 1: Foundation & Planning (Duration: 1 Week)

This initial phase is collaborative and crucial for ensuring you both build towards the same goal.

- **Joint Objectives:**
  - Research and decide on a Python backend framework (

Flask, **FastAPI**, or **Django**)[1]. FastAPI is recommended for building APIs quickly[2].

  - Set up a shared Git repository on GitHub or GitLab and ensure you are both comfortable with the basic workflow[3].
  - Thoroughly plan and define the API endpoints. This is your "contract" that details how the frontend and backend will communicate. Document what data each endpoint expects and what it will return[4].
- **Deadline Goal (End of Week 1):** Backend framework chosen, Git repo is active, and a document defining the V1 API endpoints is complete.

Phase 2: Core Component Development (Duration: 2 Weeks)

You will now work in parallel on the core parts of the application.

- **Person A (Backend & AI Logic):**
  - Set up the basic server structure using the chosen framework[5].
  - Adapt the Python code from the Colab notebook for

CV processing and structuring (handling PDF/JSON uploads and text extraction)[6666].

  - Integrate the logic for creating the

vector database from the processed CV content (embedding and storing with ChromaDB)[7777].

  - Build the first primary API endpoint (e.g.,

/analyze) that accepts a CV and a job description, and uses the retrieval logic to find relevant sections[8888].

- **Person B (Frontend & UI):**
    - Design and build the static web user interface, including the CV upload form, job link input field, and the display area for results[9].
    - Develop the frontend logic to handle file uploads and user inputs, preparing to connect to the backend[10].
- **Deadline Goal (End of Week 3):** Person A has a working /analyze endpoint that can be tested. Person B has a complete, functional frontend UI (using mock data).

Phase 3: Integration & Feature Expansion (Duration: 3 Weeks)

This phase focuses on connecting the two halves of the application and implementing the "co-pilot" features.

- **Week 4 (First Integration):**
    - **Joint Objective:** Connect the frontend to the backend /analyze API. This is a critical step that may require pair programming to resolve issues[111111].
- **Weeks 5 & 6 (Feature Expansion):**
    - **Person A (Backend):**
        - Adapt the multi-turn chat logic from the Colab notebook (Section 7) into the backend[12].
        - Create a new API endpoint (e.g.,

/chat) that takes context and a user message, and returns the AI's response[13131313].

        - Begin implementing the

job link research component that scrapes a URL and extracts the relevant text to feed into the analysis process[14141414].

    - **Person B (Frontend & Integration):**
        - Build out the dynamic chat interface on the frontend[15].
        - Connect the chat UI to the new

/chat endpoint from Person A, creating the interactive "co-pilot" experience[16].

        - Refine the overall user experience and visual design based on the integrated features[17].

- **Deadline Goal (End of Week 6):** The core application is fully integrated. A user can upload a CV, provide a job link, and have an interactive chat session with the AI to get feedback.

Phase 4: Refinement, Testing & MVP Showcase (Duration: 2 Weeks)

The final phase is about making the application robust, reliable, and ready to be shown.

- **Joint Objectives:**
    - **End-to-End Testing:** Test the entire workflow from start to finish to find and fix bugs.
    - **Refine AI Prompts:** Collaboratively refine the prompts sent to the AI to improve the quality and helpfulness of its feedback[18].
    - **Write Basic Tests:** Person B can contribute to writing tests for the backend APIs to ensure they are stable[19].
    - **Prepare for Demo:** Clean up the code, document the setup process, and prepare a showcase of your working MVP.
- **Deadline Goal (End of Week 8):** A polished and tested MVP is complete and ready to be demonstrated.